PoshanScan

Nudging Everyday Food Choices in Urban India Through a WhatsApp Based Digital Intervention

1. Abstract

India carries two nutrition problems at once. Undernutrition continues, with 35.5% of children under the age of 5 being stunted, and 57% of women being anaemic (IIPS, 2021). Along with this approximately 101 million diabetic Indians, 136 million who have prediabetes, and 40% with abdominal obesity among the adults (Anjana et al., 2023). Currently, beverages, refreshments and processed food have become the biggest shares of the total household food expenditure in both rural and urban areas (NSSO, 2024). We are concentrating on the second problem which is increasing at the fastest rate and is the more tractable target.

Indian consumers already know the risks of processed food. The hesitation comes at the time of buying. While choice architecture nudges were found to be three times more effective in averting negative choices than informational nudges in nearly 300 food choice experiments (Cadario and Chandon, 2020), Indian nutrition programmes are virtually all informational. We offer an intervention that is delivered at the point of choice, through WhatsApp: PoshanScan. Once a barcode is scanned or a shopping list is shared, the user will be given a single warning and one realistic alternative. The study will involve secondary research and will create a design that is based on evidence.

2. Introduction

2.1 Context and rationale

Food choice is a series of micro decisions made very rapidly with little thought which adds up over time to the big picture of health. According to ICMR INDIAB 17, a survey of 113,043 participants across 31 states and union territories, showed the prevalence of diabetes was 11.4% among adults, prediabetes 15.3%, hypertension 35.5%, general obesity 28.6% and abdominal

obesity 39.5% (Anjana et al., 2023). Rates are higher in urban areas and that is why we well be focusing on the urban population. Between 2005 and 2021, adult obesity and overweight rates in India grew by 91% among women (from 12.6% to 24%) and 146% among men (from 9.3% to 22.9%), making it quarter of our population (NFHS 5 IIPS, 2021).

Damage piles up in ways that are not visible. A systematic review found fatty liver disease in 38.6% of Indian adults and 35.4% of children, and in 12.4% of children who were not obese (Shalimar et al., 2022). In India, metabolic disease occurs at an earlier age and at a lower weight than in the Western world, and weight gain is not noticeable until it is too late to serve as a warning.

Ultra processed food has driven this change in diet. Its retail value in India increased by about 13% compounded between 2011 and 2021, touching 2.5 trillion rupees, with sweet biscuits about 43% of the category (WHO India and ICRIER, 2023). More than 96% of adults exceed the recommended limit of 5 g of salt per day, averaging close to 8 g per day (ICMR NIN, 2024). Affordability limits what any solution can achieve. 50% people in India, some 790 million, are unable to afford a nutritious diet (FAO et al., 2024), and only 28% of Indians eat from all five food groups daily (IFPRI, 2024). In the meantime, estimated losses due to non-communicable diseases and mental disorders reach 4.58 trillion USD by 2030 (WEF and Harvard School of Public Health, 2014).

2.2 Previous interventions

India has tried several interventions, but none targeted the moment of purchase.

The FSSAI led Eat Right India campaign promotes voluntary pledges and certification. In September 2022, FSSAI also released a draft of the Indian Nutrition Rating (FSSAI, 2022), which is star rating system which would be visible in the front of a packet. Public health researchers objected that rating systems can offset the harm of excess sugar against protein content, making poor products appear acceptable. The Supreme Court in April 2025 instructed

FSSAI to complete its recommendations on labelling within three months. In early 2026 they reported that there was no consensus among stakeholders and indicated its plans to withdraw the draft.

Commercial calorie tracking apps show low long-term engagement due to the need to create an account and manually track calories. The mDiabetes programme, sent in 12 languages via SMS for 6 months, did have an impact on behaviour, improving fruit, vegetable and fat consumption (Pfammatter et al., 2016). Its strength was that it used a channel that people already had. These examples illustrate the problem with the existing approach. Even simple and affordable solutions underperform because they focus on information instead of on the customer at the point of sale.

3. Literature Review

3.1 Behavioural economics

A nudge alters the temptation or incentive, but not through a ban or a tax (Thaler and Sunstein, 2008). Cadario and Chandon (2020) reviewed almost 300 food choice experiments, reporting that on average, altering the presentation of an option is three times as effective as informational intervention. It is better to change the default than to ask people to change, and it is better to change the way unhealthy options are presented than to emphasise the benefits of healthy options.

Researchers have scrutinised these findings closely. Mertens et al. (2022) reported nudging to be generally effective across various domains, but a reanalysis by Maier et al. (2022) found that once publication bias is corrected, the average effect largely disappears. Nudging is therefore a real but bounded instrument, and we do not claim otherwise.

3.2 Digital nutrition interventions

Kurniawan et al. (2025) reviewed 53 digital interventions which targeted diet and physical activity in healthy populations in low and middle income countries through social media, SMS,

apps, games and websites. There were positive changes in knowledge, attitudes and self-reported behaviour, and mixed changes in body measurements.

Behaviour change theory-based interventions were more likely to be effective, and there was no difference between multi-channel and single channel programmes. Users disengage, and this remains the main obstacle. Eysenbach (2005) reported that most participants drop out within the first few weeks, and a meta analysis of app based chronic disease interventions found that the average drop out rate was 43% (Meyerowitz Katz et al., 2020).

3.3 Evidence for labelling interventions

Warning labels are one of the most promising directions. The law introduced in Chile, which includes black octagon warning labels as well as marketing restrictions, led to a reduction in the sales of sugary drinks of 23.7% (Taillie et al., 2020). A warning label summarises the information and is more effective than a nutrition table. We borrow that visual logic rather than the policy itself, since front of pack labelling in India remains unresolved.

3.4 Point of purchase interventions

Several point-of-sale experiments demonstrate what works. Stuber et al. (2022) found that informational nudges raised healthy purchases by around 2.4% in a Dutch online supermarket with 11,775 participants, with the effect concentrated among lower income shoppers. In a UK online supermarket, a change of default led to a reduction of around 5% of saturated fat in the basket, while the alternatives provided reduced it by about 2% (Koutoukidis et al., 2019). Wyse et al. (2021) ran the Click and Crunch trials in Australian school canteens, combining labelling, defaults, prompts and feedback, and recorded almost a ten per cent greater uptake of healthy food options after 12 months. Defaults worked better than suggestions, and combinations worked better than single tweaks. This finding also establishes a realistic expectation, since the gains were measured in single digit percentages rather than dramatic change.

3.5 Food environment in India

Indian grocery shopping is becoming digital, with quick commerce platforms favouring snack items and beverages because rapid delivery rewards impulse. In this context, choice architecture has an immediate relevance, since the consumer interaction with the product already occurs on a screen.

In a study of 1,831 consumers in supermarkets in Delhi and Hyderabad, about 90% of consumers said they read labels, but 81% checked only the manufacturing or expiry date and just 33% checked the nutrition panel (Vemula et al., 2014). The reasons mentioned were small font size, technical language and time pressure. Meanwhile WhatsApp leads with more than 500 million Indian users, and a new nutrition app is unlikely to be adopted. Social factors are also important, as food plays a key role in the life of the Indian family and purchasing decisions are generally not made by a single individual.

3.6 Gaps, needs and risks

The review points to several gaps. Most nutrition interventions in India are geared towards knowledge rather than towards the actual purchase, so they are informative but not transformative. Second, we are not aware of any study which has looked at the impact of choice architecture on food purchasing in India, although quick commerce is a fast growing trend. Third, few interventions operate at the point of sale, and those that do have been limited by the lack of regulatory support following the stalling of front of pack labelling.

Method and equity raise further problems. Most interventions carry the problem of recall bias, so objective purchase measures are needed even though they are harder to obtain. Affordability is a serious constraint, since recommending expensive alternatives to cheap processed food is counterproductive for households already unable to afford a healthy diet. Most interventions also assume a user who reads English and is comfortable with technology, which describes a minority of the country. Literacy and device access vary too widely for that assumption to hold.

One further complication is that in many households the person who buys the food does not control the money, so a persuaded user may still lack on the authority to act on the nudge

Our own limits are worth stating plainly. Our work rests on published literature, some of it behind paywalls we cannot get past. That means our recommendations come from what others have tested, not from anything we have tested ourselves. Where a figure comes from an industry source rather than a peer reviewed one, we say so.

4. Project Objectives

1. To assess the prevalence of diet related lifestyle disease among urban Indian consumers and to understand the purchasing factors driving it, using secondary data.
2. To identify the behavioural and informational obstacles preventing the use of nutritional information to achieve better health outcomes.
3. To develop a nutrition nudge, called PoshanScan, based on a named behaviour change mechanism and on literature evidence.
4. To assess the feasibility and ethics of the intervention, including retention, availability of substitutes and reputational risk.

5. Methodology

We use secondary sources, since the research has a time limit and a great deal of published material is already available. The review is based on literature with India specific information on dietary change, supported by international trials where no India specific evidence exists.

5.1 Sources

We will draw on the following sources:

- Government reports. The International Institute for Population Sciences, the Household Consumption Expenditure Survey and the National NCD Monitoring Survey.
- ICMR publications. The INDIAB series, the ICMR NIN Dietary Guidelines for Indians 2024 and National Centre for Disease Informatics material.

- FSSAI publications. Labelling notifications, the Indian Nutrition Rating and Eat Right India material.
- Research articles. PubMed, PubMed Central, ScienceDirect, SpringerLink, Lancet journals, PLOS Medicine and the Journal of Medical Internet Research.
- International organisations. The World Health Organization India country office, FAO, the World Bank and the World Economic Forum.
- Policy think tanks. ICRIER, NITI Aayog and the International Food Policy Research Institute.
- Regulatory material. Department of Consumer Affairs guidelines on deceptive design and the Digital Personal Data Protection Act 2023.

6. Tangible Solution and Output

6.1 Description of the intervention

PoshanScan is a nutrition nudge that is sent using WhatsApp. There is no need to install an application, create a profile or log meals. The user takes a picture of a barcode or a list of purchases and receives two things in reply. First a warning, which categorises the product and shows visually how it compares against the daily guidance issued by ICMR NIN. Second a substitution, offering one simple and feasible alternative.

6.2 Justification

We chose WhatsApp because the literature supports it. Standalone nutrition apps lose users, with average attrition of 43% in chronic condition management (Meyerowitz Katz et al., 2020), whereas messaging through an existing app is easier to sustain and has been popular with Indian consumers (Pfammatter et al., 2016). Additionally, the intervention is targeted at the point of purchase, as Kurniawan et al. (2025) suggest that information provided afterwards has a smaller effect while information provided during the decision has measurable effects.

Behavioural economics informs the design. It works on the principles of commitment and consistency, since a consumer who is presented with a signal is more likely to take action, and it reduces decision fatigue by offering one option rather than a list. Changing the default is the most effective lever, but it is not available to us as a standalone tool, according to Cadario and Chandon (2020).

We also designed around the barriers to engagement. Most Indian consumers do not read nutrition labels, so the intervention supplements the existing information rather than replacing it, for example, by quantities are given in terms that can be related to the reader, such as teaspoons of sugar

6.3 Deliverable outputs

The following deliverables will be made available for proof of concept purposes.

1. A document that summarises the intervention, its benefits and its limitations.
2. Message concepts for PoshanScan in natural language.
3. A table of supporting evidence with annotations.
4. A feasibility and ethics study covering retention, affordability of substitution, compliance with the Digital Personal Data Protection Act 2023 and deceptive design guidelines.

7. Contribution to the UN Sustainable Development Goals

This project supports Sustainable Development Goal 3 on good health and wellbeing, particularly target 3.4 on reducing mortality from non-communicable diseases.. In India, diet is among the most important risk factors for diabetes, hypertension and cardiovascular disease, and this intervention addresses that root cause by focusing on how food is purchased. It also supports 12.8 of Goal 12 on enabling informed choices. To a lesser degree it contributes to Goal 10 on reduced inequalities, as it is developed for users who lack high literacy skills and uses technology they already own.

8. Project Timeline

This project will be completed over three months. The secondary source research, the extraction of statistics with citations and the gap analysis form the first stage, which ends in the first week of September. The second phase, ending in the first week of October, consists of developing the prototype of the message flow for PoshanScan, namely the scan, the warning and the substitution suggestion. The third phase completes the software and the report, including the evidence table and the full ethics review, and is due by the end of the second week of November.

9. Disciplinary Perspective

PoshanScan is interdisciplinary and none of the disciplines alone could deliver it. It is for this reason that our team is organised the way it is. Economics provides the theoretical backbone of the project, since nudge theory and choice architecture come from behavioural economics, and it also challenges us to examine the affordability constraint identified in our own review, which is the single biggest limit on any dietary change in India. From marketing we take an account of why processed food becomes so popular, along with the skill of writing messages that persuade without blaming the user, which matters because a shaming tool will be shut down. Whether the intervention is usable at all is a question of design, since most Indian shoppers skip nutrition labels because of small print and technical terms, so the information must be unambiguous and accessible to someone with limited literacy.

Computer science brings the system to life through the WhatsApp Business API integration, the scanning feature and the backend that delivers a response at the point of sale. International relations places our work in the global policy debate, drawing on the labelling regimes in Chile and Mexico and the current discussion about front of pack labelling in India. Business analytics and finance support the evaluation, allowing purchase data to become a more stringent metric than reported diet and testing whether the tool could be sustained past a pilot. This is not simply

a convenient way to present work across disciplines. What we discovered was that most nutrition tools created in India are created by one type of expert and therefore fail on some other dimension, and our composition is a direct response to that.

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